

APPLYING A MULTI-LAYER PERCEPTRON (MLP) MODEL ON THE MNIST DATASET

1-DATASET IDENTIFICATION AND PRELIMINARY RESEARCH

The chosen dataset for this study is the MNIST dataset. This dataset comprises images of handwritten digits from 0 to 9, a training set of 60,000 images, and a test set of 10,000 images. The dataset and associated code were sourced from Yann LeCun's website, the creator of the dataset.

2- MODEL SELECTION AND PREPARATION

The Multi-Layer Perceptron (MLP) model, a form of neural network, was chosen for classifying the images. An appropriate pretrained MLP model was obtained from Kaggle, a platform that provides various pretrained models.

3- IMPLEMENTATION

Dataset Setup:

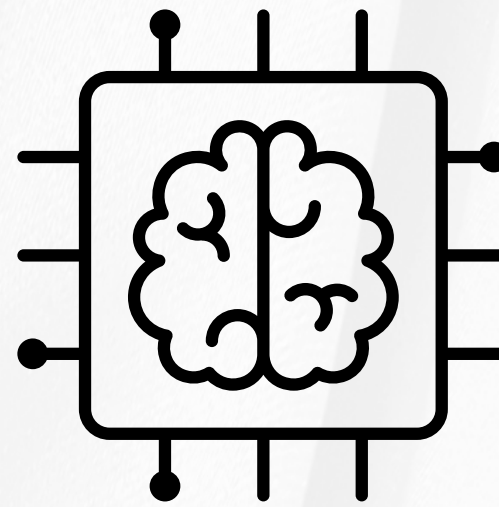
The dataset was downloaded and stored in the same directory as the script. The script was then executed using the R console.

Model Implementation:

The provided script aims to build a neural network in R. It takes the dataset as an input, pre-processes it, and utilizes it to train the neural network. The script also includes functions to test the accuracy of the neural network and visualize the output of the network.

Model Training:

A neural network with two layers was built. The network uses ReLU as the activation function and softmax as the cost function. It is trained using gradient descent with a specified learning rate and regularization strength.



AT A GLANCE

This case study illustrates the feasibility of applying an MLP model on the MNIST dataset, indicating a practical example of digit recognition. Future work might involve experimenting with different network architectures, activation functions, and regularization techniques to further optimize model performance.

MODEL TESTING AND EVALUATION

The model was tested using both the spiral dataset and the MNIST dataset. The output of the code is the loss function value, which is presented every 1000 iterations. The aim of training the network is to minimize this loss function value, demonstrating the effectiveness of the network at classifying the dataset.